

Experiment 20 — Future Sales Prediction

AIM

To implement a Future Sales Prediction model using a suitable machine learning algorithm in Google Colab.

SOFTWARE USED

Google Colab

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required libraries.
3. Create a historical sales dataset containing month and sales values.
4. Separate the month as the input feature and sales as the target variable.
5. Split the data into training and testing sets.
6. Create a Linear Regression model.
7. Train the model using historical sales data.
8. Predict sales for the test data.
9. Evaluate the model using Mean Squared Error (MSE) and R^2 Score.
10. Use the trained model to predict sales for a future month.
11. Visualize the historical and predicted sales.

CODE

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
# Create historical sales dataset
```

```
data = {
    "Month": list(range(1, 13)),
    "Sales": [120, 135, 150, 160, 175, 190,
              205, 220, 235, 250, 270, 290]
}
df = pd.DataFrame(data)
print("Historical Sales Data:")
print(df)
# Input and target
X = df[["Month"]]
y = df["Sales"]
# Create Linear Regression model
model = LinearRegression()
# Train model using historical data
model.fit(X, y)
# Predict historical sales
y_pred = model.predict(X)
# Evaluate model
mse = mean_squared_error(y, y_pred)
r2 = r2_score(y, y_pred)
print("\nMean Squared Error:", mse)
print("R2 Score:", r2)
# Predict future sales
future_month = [[13]]
future_sales = model.predict(future_month)
print("\nPredicted Sales for Month 13:", future_sales[0])
# Visualization
plt.figure(figsize=(8, 5))
plt.scatter(
```

```

    df["Month"],
    df["Sales"],
    label="Actual Sales"
)
plt.plot(
    df["Month"],
    y_pred,
    label="Regression Line"
)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Future Sales Prediction")
plt.legend()
plt.show()

```

OUTPUT

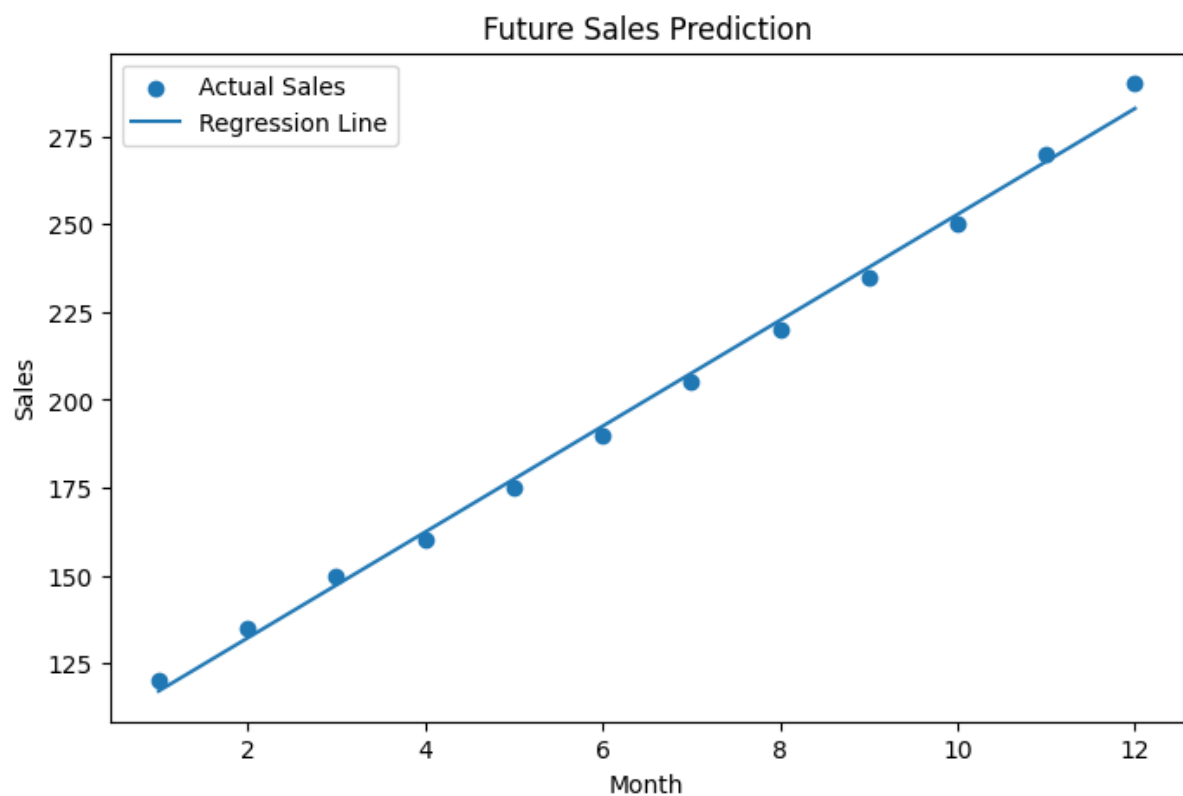
Historical Sales Data:

	Month	Sales
0	1	120
1	2	135
2	3	150
3	4	160
4	5	175
5	6	190
6	7	205
7	8	220
8	9	235
9	10	250
10	11	270
11	12	290

Mean Squared Error: 10.358391608391573

R2 Score: 0.9961870951134755

Predicted Sales for Month 13: 297.9545454545454



RESULT

Thus, the Future Sales Prediction model was successfully implemented using Linear Regression in Google Colab, and future sales were predicted based on historical sales data.